Reasoning over User Preferences: Knowledge Graph-Augmented LLMs for Explainable Conversational Recommendations

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***Abstract*—Conversational Recommender Systems (CRSs) aim to provide personalized recommendations by capturing user preferences through interactive dialogues. Explainability in CRSs is crucial as it enables users to understand the reasoning behind recommendations, increasing system transparency and trustworthiness. However, current CRSs often leverage knowledge graphs (KGs) or language models to extract and represent user preferences as latent vectors, which limits their explainability. Large language models (LLMs) offer powerful reasoning capabilities that can bridge this gap by generating human-understandable preference summaries. However, effectively reasoning over user preferences in CRSs remains challenging as LLMs pre-trained on large-scale corpora may not be well-suited for analyzing user preferences, which requires domain-specific knowledge. While KGs provide rich domain knowledge, integrating them with LLMs encounters a significant modality gap between structured KG information and unstructured conversations. In this paper, we propose COMPASS (Compact Preference Analyzer and Summarization System), a plug-and-play framework that synergizes LLMs and KGs to reason over user preferences, enhancing the performance and explainability of existing CRSs. COMPASS employs a two-stage training approach: first, it bridges the gap between the structured KG and natural language through novel graph entity captioning pre-training. This enables the LLM to transform KG entities into natural language descriptions, allowing it to comprehend domain-specific knowledge. Next, COMPASS optimizes user preference reasoning via knowledge-aware instruction fine-tuning, where the LLM learns to reason and summarize user preferences from dialogue histories and KG-augmented context. This enables COMPASS to perform knowledge-aware reasoning and generate interpretable user preferences that can seamlessly integrate with existing CRS models for improving recommendation performance and explainability. Our experiments on benchmark datasets demonstrate the effectiveness of COMPASS in improving various CRS models.**

***Index Terms*—Conversational Recommender System, Large Language Model, Knowledge Graph, Explainable Recommendation**

I. Introduction

Providing personalized, context-aware recommendations that align with users' changing preferences and situational needs is a critical challenge across various domains such as e-commerce, streaming platforms, and other online services. In recent years, conversational recommender systems (CRSs) have emerged as a promising approach, harnessing the power of natural language interactions to unravel user preferences [1], [2]. By engaging the user in interactive conversations, CRSs enable a better understanding of the user's evolving interests and guide them toward products, services, and information that best meet their immediate requirements [3], [4].

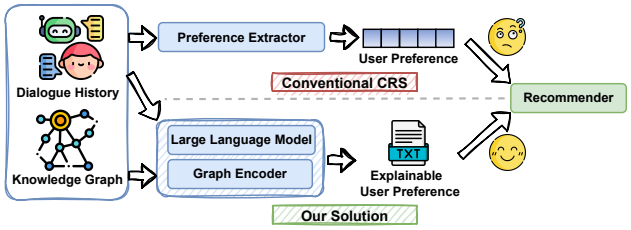

Fig. 1: Conventional CRSs lack explainability in user preference modeling by extracting hidden representation, while our approach enhances transparency by generating interpretable user preference in text.

Despite the success, CRSs face unique challenges in precisely modeling user preferences from semantically rich and dynamically evolving dialogue. They must interpret often ambiguous and context-dependent natural language inputs [4], [5], while continuously updating and refining their understanding of user preferences in real-time [6], integrating immediate interests with overall preferences developed over time.

Conventional CRSs mainly depends on item-centric approaches [5], [7], focusing on explicitly mentioned items in the conversation to model user preference. Recent studies incorporating pre-trained language models (PLMs) to enhance both natural language understanding and user preference modeling [8], [9]. However, these methods often fail to infer implicit preferences or reason about underlying motivations beyond explicit item mentions, leading to a superficial understanding of user intent. Moreover, they typically represent user preferences as hidden embeddings, leaving it unclear what specific preference is being considered when making a recommendation, as shown at the top of Figure 1. This ambiguity makes it challenging to verify the underlying reasons and results behind the recommendations, hindering the transparency and accountability of the recommender system.

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Recent advancements in large language models (LLMs), have demonstrated exceptional capabilities in both language understanding [10], [11] and complex reasoning [12]. These capabilities make LLMs promising for explainable recommendations, and recent works [13]–[15] have demonstrated the LLM's ability to reason over user historical behaviors and generate interpretable recommendation explanations. Nevertheless, these methods only focus on the traditional recommendation settings, where user preferences are static. In contrast, CRSs require more sophisticated reasoning to model user preferences that evolve dynamically throughout conversation. While recent work has explored LLMs for various CRS tasks, including evaluation [16], [17], zero-shot recommendation [18], and task planning [19], [20], the role of explainable user preferences in LLM-based CRSs remains unexplored.

Enabling LLMs to reason over user preferences requires more than their general language capabilities. Effective preference reasoning requires domain-specific knowledge about item attributes, relationships, and up-to-date information that LLMs inherently lack. Knowledge graphs (KGs) provide this essential structured knowledge through rich semantic relationship and have proven effective in CRSs [5], [7], [8], [21]. Despite their demonstrated potential, integrating KGs with LLMs for preference reasoning presents fundamental challenges that existing approaches have not addressed:

- **Modality Gap (Challenge 1):** There exists a significant modality gap between KGs and LLMs. While LLMs process sequences of tokens representing natural language, KGs represent information in structured, graph-based formats with explicit entity relationships. This representational mismatch limits LLMs's ability to directly interpret and reason with the domain knowledge encoded in KGs.
- **Cross Modal Reasoning (Challenge 2):** Effectively reasoning over both the KG and conversation to infer user preferences is a complex task. LLMs, despite their strong natural language processing capabilities, are not inherently designed to perform this cross-modal reasoning. Therefore, they face difficulties in analyzing and synthesizing insights from graph-structured knowledge alongside dialogue history. This limitation hinders their ability to identify relevant patterns across both sources and to perform the knowledge-aware reasoning necessary for comprehensive user preference modeling.

To address the above challenges, we propose **Com**pact **P**reference **A**nalyzer and **S**ummarization **S**ystem (COMPASS), a novel framework that augments the LLM with the KG to enable explainable preference reasoning, improving both recommendation performance and the explainability of existing CRSs. COMPASS introduces a two-stage training paradigm that fundamentally transforms how LLMs process and reason with structured knowledge for preference analysis. First, we introduce *graph entity captioning pre-training* that transforms KG structures into natural language descriptions. This allows the LLM to comprehend domain-specific information and bridge the **modality gap (Challenge 1)**. We employ a Graph Neural Network (GNN) to encode structural information from the KG as entity embeddings, which are then projected into the LLM's semantic space via an adapter module. The LLM learns to decode these projected embeddings into textual descriptions of entities and their relationships, effectively translating graph-structured knowledge into a format compatible with its reasoning mechanisms. Building upon this alignment, COMPASS employs *knowledge-aware instruction fine-tuning* to enhance the LLM's ability to reason about user preferences from dialogue histories and KG-augmented contexts. These KG-augmented contexts consist of relevant entity information and relationships extracted from the KG, providing a rich background for inference beyond the conversation history alone. Through carefully designed instructions, we enhance the LLM's capability to perform **cross-modal reasoning (Challenge 2)** by analyzing conversation history and cross-referencing with KG-augmented information. This instruction tuning process enhances the LLM's ability to extract explicit mentions, infer implicit interests, and reason about preferences in relation to various item attributes. Consequently, as shown in the bottom of Figure 1, COMPASS generates comprehensive and interpretable user preferences in text that capture both overall preferences and current interests. To leverage these insights, we introduce an adaptive gating mechanism that integrates summarized preferences into existing CRSs, boosting recommendation performance and explainability without requiring architectural changes. Our main contributions can be summarized as follows:

- **New framework.** We propose COMPASS, a novel framework for enhancing user preference modeling in CRSs. To the best of our knowledge, this is the first work to leverage LLMs and KGs for explainable preference generation in CRSs.
- **Effective cross-model reasoning and explanation.** We develop a two-stage process that enables the LLM to perform cross-modal reasoning over KGs and conversations, generating explainable user preference summaries. This approach moves beyond abstract vector representations to provide clear, human-readable user preferences.
- **Flexible plug-in.** COMPASS generates user preference summaries that are compatible with existing CRS architectures without requiring modifications to the system, improving both recommendation performance and explainability.

II. Related Work

Knowledge graphs enable systems to infer implicit user interests and capture item relationships [5], [7], **Conversational Recommender System.** Conversational recommender systems (CRSs) understand user preferences through interactive dialogue to provide personalized recommendations [1]. These systems face a unique challenge in modeling user preferences from typically brief and sparse conversational data. To address this, recent works incorporate external knowledge sources into CRSs to enhance preference modeling. Knowledge graphs

enable systems to infer implicit user interests and capture item relationships [5], [7],, while user reviews provide additional insights into item attributes and user opinions [22], [23], helping to address the challenge of limited contextual information. Moreover, pre-trained language models like DialoGPT [8], [9] and BART [21], [24] has further improved contextual understanding and language generation capabilities in CRSs.

More recently, the advent of powerful LLMs has introduced new opportunities for CRSs. Studies have focused on improving LLM evaluation through enhanced user simulators [16], [17], exploring zero-shot recommendation capabilities [18], using graph-based retrieval augmentation for in-context learning [25], and utilizing LLMs for task planning [19], [20]. However, one critical aspect that remains underexplored is the role of explainable user preference in CRSs. Although some conceptual designs propose using LLMs to manage user preferences [26], they lack quantitative validation. MemoCRS [27] addresses user preference memories in sequential CRS but does not fully harness LLM's reasoning abilities for preference generation. Our work introduces the first framework that augments LLMs with KG for reasoning over user preferences in CRSs, enhancing both preference modeling and recommendation explainability. Our work introduces the first framework that augments LLMs with KG for reasoning over user preferences in CRSs, enhancing both preference modeling and recommendation explainability.

Explainable Recommendation. Explainable recommendation aims to enhance user trust and decision-making by providing transparent rationales for recommendations. Early research leveraged user-item interactions and content features to generate simple explanations [28], using techniques like recurrent neural networks [29], attention mechanisms [30], and graph neural networks [31]. Recent developments in LLMs have opened new avenues for explainable recommendation. Recent LLMs have shown potential in generating explanations by analyzing user interactions [13], [14], but primarily operate within static user profiles where preferences are inferred from historical behavior. Our work introduces a novel KG-augmented LLM framework for CRSs, enhancing both preference modeling and recommendation explainability in dynamic interaction contexts.

III. Preliminaries

Conversational Recommendation. Let i denote a candidate item from the set of items $\mathcal{I}$, and let w denote a word in the vocabulary $\mathcal{V}$. A dialogue $\mathcal{D}$ between a user and the recommender system consists of a sequence of utterances $\mathcal{D} = [u_1, \ldots, u_T]$, where $u_t = [w_1^t, \ldots, w_m^t]$ is the t-th utterance composed of m words, and T is the maximum number of turns in the dialogue. As the conversation progresses, the dialogue history up to the turn t is denoted as $H_t = [u_1 : u_t]$, where $[u_1 : u_t]$ signifies the chronological sequence of utterances from the first to the t-th turn. The CRS estimates the user's preferences based on H_t, then recommend K items from $\mathcal{I}$, which are used to generate the next utterance u_{t+1}. Note that $\mathcal{I}_t$ can be empty when no recommendation is needed. In such cases, the CRS may raise a clarification question or generate a casual conversation response.

Knowledge Graph. A knowledge graph is defined as $\mathcal{G} = (\mathcal{E}, \mathcal{A}, \mathcal{X})$ where $\mathcal{E}, \mathcal{R}$ represents the set of entities and relation types in the graph, respectively. $\mathcal{A}$ is the adjacency matrix capturing the relationships between entities and $\mathcal{X}$ represents the textual descriptions of each entity. For each entity $e \in \mathcal{E}$, its description is denoted as $x_e \in \mathcal{X}$, where $x_e = [w_1^e, \ldots, w_k^e]$, and w_k^e represents the k-th word in the entity description. The entity set $\mathcal{E}$ encompasses candidate items $\mathcal{I}$ (e.g., movies) and non-item entities that represent item attributes (e.g., actors, genres, keywords). Formally, $\mathcal{I} \subseteq \mathcal{E}$.

Explainable User Preferences and Recommendations. Explainable user preferences are crucial for enhancing the transparency and effectiveness of CRSs. Our goal is to generate clear, human-understandable textual user preference summaries that provide insights for both recommendation and explanation. Specifically, for a given dialogue history H_t, we define the generation of user preference summaries as:

$$\mathcal{P}_t = f(I_p, H_t, \mathcal{E}_t^m, \mathcal{G}), \tag{1}$$

where f represents a model that reasons over the dialogue history H_t, the mentioned entities $\mathcal{E}_t^m \subseteq \mathcal{E}$, and their associated information from the knowledge graph $\mathcal{G}$. I_p is denoted as instruction prompts. The resulting $\mathcal{P}_t$ represents the textual explainable user preference summary at the t-th turn. The summary $\mathcal{P}_t$ is then encoded using a preference encoder $g(\cdot)$ and integrated into the base CRS model f_{crs}, which uses it to adjust its recommendation strategy. Depending on the model, additional inputs, such as KG information or dialogue history, may also be used. Formally, the recommendation step is represented as:

$$\mathcal{I}_t = f_{\text{crs}}(g(\mathcal{P}_t), H_t, \mathcal{G}), \tag{2}$$

where $\mathcal{I}_t$ represents the recommended items at turn t, and $g(\mathcal{P}_t)$ denotes the encoded preference summary.

IV. Methodology

A. Overview

The primary goal of COMPASS is to synergize the reasoning capabilities of LLMs with the structured knowledge from KGs to analyze and summarize user preferences. COMPASS comprises three core components: (1) *Graph encoder* processes a domain-specific KG, capturing complex relationships between items to augment user preference modeling. (2) *Graph-to-Text adapter* aims to bridge the modality gap between the graph encoder and the LLM, enabling the LLM to comprehend the graph structure and conduct reasoning. (3) *Large Language Model (LLM)* leverages the powerful reasoning and generative capabilities of advanced language models to generate interpretable user preference summaries.

To integrate the components cohesively, we employ a two-stage training process: (1) *Graph entity captioning* aligns KG structures with natural language representations, creating a shared semantic space for the LLM to comprehend and reason with domain-specific knowledge effectively. (2) *Knowledge-aware instruction fine-tuning* optimizes the LLM for cross-

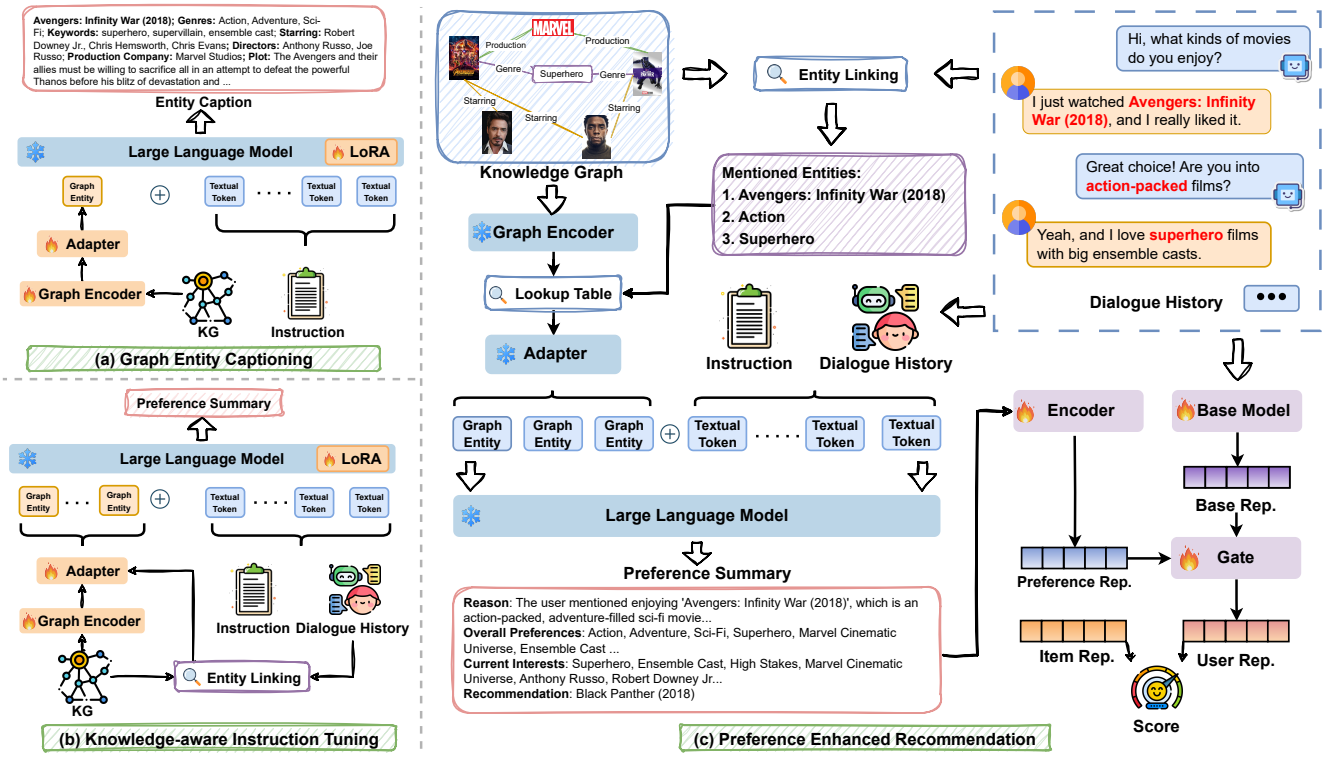


Fig. 2: The overall framework of our COMPASS. COMPASS consists of three components: the graph encoder, the adapter, and the LLM. The adapter aligns the knowledge graph to the LLM. COMPASS follows a two-stage training paradigm - (a) *Graph Entity Captioning* and (b) *Knowledge-aware Instruction Tuning*. Once trained, COMPASS can be easily integrated with existing CRS models for *(c) Preference Enhanced Recommendation*.

modal reasoning, allowing it to generate comprehensive user preference summaries by synthesizing information from dialogue history and KG-augmented context. Once trained, COMPASS can be integrated with existing CRS models to improve their recommendation performance and explainability by generating user preferences and incorporating them into the recommendation process with an adaptive gating mechanism. Figure 2 illustrates COMPASS's architecture and training process.

B. Model Architecture

1) Graph Encoder: In COMPASS, the KG is a crucial source of domain-specific information that provides extra context for understanding attributes and relationships of items. To efficiently convert the structured knowledge in the KG into a format understandable by the LLM for preference reasoning and summarization, we utilize a Relational Graph Convolutional Network (R-GCN) [32] to capture the complex graph structure and generate entity embeddings. The R-GCN is well-suited for modeling KGs due to its ability to handle multi-relational data and capture higher-order dependencies between entities.

To initialize the entity embeddings, we leverage the textual descriptions $\mathcal{X}$ associated with the entities in the KG. Specifically, for each entity $e \in \mathcal{E}$, we encode its description x_e using a PLM. This provides a rich semantic foundation for the graph learning process. The R-GCN then captures both entity-level information and the overall graph structure through iterative message passing, which is particularly important for understanding the relationships between items and attributes. Formally, the representation of an entity e at the l-th layer is calculated as follows:

$$\mathbf{h}_e^{(0)} = \text{PLM}(x_e), \tag{3}$$

$$\mathbf{h}_e^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{e'\in\mathcal{E}_e^r}\frac{1}{Z_{e,r}}\mathbf{W}_r^{(l)}\mathbf{h}_{e'}^{(l)} + \mathbf{W}_e^{(l)}\mathbf{h}_e^{(l)}\right), \tag{4}$$

where $\mathbf{h}_e^l$ is the embedding of entity e at the l-th layer, $\mathcal{E}_e^r$ is the set of neighboring entities connected to e through relation r, $\mathbf{W}_r^{(l)}$ and $\mathbf{W}_e^{(l)}$ are learnable weight matrices, $Z_{e,r}$ is a normalization factor, and σ is an activation function.

2) Graph-to-Text Adapter: The entity embeddings from the graph encoder exist in a different representational space from the LLM's textual token representations. This makes it challenging for the LLM to effectively reason with the KG-augmented context to generate user preferences. To bridge this semantic gap and enable effective knowledge integration, we introduce an adapter module that creates a mapping between graph-structured entity embeddings and the LLM's textual domain. Specifically, the adaptation process is defined as:

$$\mathbf{h}_e^{\tau} = f_P(\mathbf{h}_e), \tag{5}$$

where $\mathbf{h}_e^\tau$ is the adapted entity embedding aligned with the LLM's semantic space, $\mathbf{h}_e$ is the entity embedding generated from graph encoder, and f_P is the projection function implemented as a linear layer.

3) Large Language Model (LLM): The LLM is the main reasoning engine that generates user preferences by synthesizing information from dialogue histories and KG-augmented contexts. It captures both the explicit user preferences expressed in the dialogue and the implicit preferences inferred from the KG-augmented context for accurate preference reasoning. Our framework is compatible with various state-of-the-art LLMs, allowing flexibility in model choice.

C. Training Pipeline

1) Graph Entity Captioning: To obtain a Graph-to-Text adapter that bridges the modality gap between the graph entity embeddings and the LLM's semantic space, inspired by the pre-training strategies employed in vision-language models [33], [34], we introduce a graph entity captioning pre-training mechanism, as shown in Figure 2a. This process creates a strong connection between graph-structured data and natural language by generating entity-specific captions that combine intrinsic entity information with neighboring node data, simulating message-passing within the graph structure. The pre-training stage enables the LLM to interpret graph entity embeddings in a semantic context, facilitating deeper understanding of KG relationships and attributes for preference modeling.

The caption generation process differentiates between item entities (e.g., movies) and non-item entities (e.g., genres, directors). For item entities, we employ a structured template that captures key attributes and relationships.

Example of Item Entity Caption Template

Movie Title: *<Title>;* ***Genres:*** *<Genres>;* ***Keywords:*** *<Keywords>;* ***Starring:*** *<Actors>;* ***Directors:*** *<Directors>;* ***Production Company:*** *<Company>;* ***Plot:*** *<Plot Summary>*

This template ensures comprehensive coverage of item attributes while maintaining a consistent structure across different items. For non-item entities, we adopt a more flexible approach that emphasizes the entity's role and its connections within the KG.

To format the training data, an input-output pair is constructed for each entity in the knowledge graph. The input consists of the adapted entity embedding $\mathbf{h}_e^\tau$ from Equation 5 and a task-specific instruction prompt I_c. The output is the generated caption C_e for the entity e. This format is represented as follows.

Entity Captioning

Input: <Entity Embeddings $\mathbf{h}_e^\tau$ >, <Instruction I_c>
Output: <Entity Caption C_e>

In this way, the LLM learns to map the graph-structured input to natural language by reconstructing the entity caption, conditioned on the graph entity embeddings and instructions. This process is optimized by minimizing the negative log-likelihood (NLL) of the generated captions, as expressed by:

$$\mathcal{L}_{caption} = -\sum_{e \in \mathcal{E}} \log P(C_e \mid \mathbf{h}_e^\tau, I_c), \tag{6}$$

where $\mathcal{E}$ is the set of all entities in the KG. Through the training, the LLM learns to interpret graph entity embeddings in a semantic context, leading to a deeper understanding of the relationships and attributes encoded within the KG. This understanding is essential for improving subsequent preference modeling.

2) Knowledge-aware Instruction Tuning: After the graph entity captioning pre-training, the LLM has gained a basic understanding of the KG structure and content. However, it has not yet been explicitly trained to utilize this knowledge for downstream tasks such as preference modeling and recommendation generation. To this end, we introduce knowledge-aware instruction tuning to enable the LLM to reason across modalities, integrating dialogue history and KG information to infer user preferences and interests, as shown in Figure 2b.

The knowledge-aware tuning process employs a carefully crafted instruction prompt I_p to guide the LLM in synthesizing and reasoning over inputs from multiple sources. Given a dialogue history H_t and the entities $\mathcal{E}_t^m$ mentioned within it, the process retrieves the embeddings $\mathbf{E}_t$ of these entities from Equation 5. These embeddings, along with the full dialogue history, serve as inputs for the LLM. The instruction prompt I_p directs the LLM to analyze both the KG-derived entity information and the dialogue history to generate a user preference summary $\mathcal{P}_t$.

This prompt follows a coarse-to-fine structure containing four key steps: (1) reasoning, providing transparency in the model's decision-making; (2) overall preferences, offering a broad view of the user's tastes; (3) current interests, capturing recent and specific preferences to guide subsequent recommendations; and (4) recommendation, leveraging the LLM's reasoning capabilities to suggest relevant items aligned with user preferences, guiding downstream CRS models. This structured prompt ensures that the model captures both long-term preferences and immediate interests. To generate ground-truth preference summaries, we utilize an advanced LLM (e.g., ChatGPT) that performs cross-modal reasoning, integrating complete dialogue histories with structured metadata of mentioned items from the KG. The instruction tuning process can be represented in the following format:

Knowledge-aware Instruction Tuning

Input: <Mentioned Entities Embeddings $\mathbf{E}_t$>, <Instruction I_p>, <Dialogue History H_t>
Output: <Preference Summary $\mathcal{P}_t$>

A detailed example of the preference summary can be found in Table IV. This instruction tuning process is optimized by

minimizing the NLL of the generated preference summaries, as expressed by:

$$\mathcal{L}_{preference} = -\sum_{\mathcal{D}\in\mathcal{C}} \sum_{H_t\in\mathcal{D}} \log P(\mathcal{P}_t \mid \mathbf{E}_t, I_p, H_t), \quad (7)$$

where $\mathcal{C}$ represents the set of all dialogues in the training data. Through this process, COMPASS learns to synthesize information from dialogue history and KG-derived entity embeddings, enabling it to generate comprehensive and interpretable preference summaries.

D. Integration with Existing CRS Models

To enhance the existing CRS models with the user preference summaries generated by COMPASS, we propose a two-step integration process: (1) transforming the natural language preference summaries into a format compatible with CRS models, and (2) incorporating these transformed preferences to enhance the base CRS model's recommendation performance via an adaptive gating mechanism. Note that COMPASS remains frozen during this process.

1) Preference Representation: Traditional CRSs are not designed to directly utilize natural language preference summaries. We employ a PLM-based encoder to transform these summaries into a format suitable for existing CRS architectures. The encoder is adaptable and can be implemented as either a frozen or trainable text encoding model. We employ BERT [35] to encode the user preference summary as follows:

$$\boldsymbol{s}_c = \text{PLM}(\mathcal{P}), \quad (8)$$

where $\boldsymbol{s}_c$ is the encoded preference from text preference $\mathcal{P}$, specifically the [CLS] token embedding. The contextual understanding of language models enables $\boldsymbol{s}_c$ to capture comprehensive user preference summaries at both coarse-grained (i.e., overall user preferences) and fine-grained levels (i.e., current interests and specific items).

2) Enhanced Recommendation: We integrate the encoded preferences into existing CRS models to enhance their recommendation performance. We employ an adaptive gating mechanism to enhance the preference representation $\boldsymbol{s}_b$ captured from the base CRS model [1] with our COMPASS-generated representation $\boldsymbol{s}_c$. Note that the preference representation $\boldsymbol{s}_b$ is specific to each CRS model as different models employ distinct methods for building user preferences. Formally, we have:

$$\gamma = \sigma(\mathbf{W}[\boldsymbol{s}_b; \boldsymbol{s}_c]), \quad (9)$$

$$\boldsymbol{s}_u = \gamma \boldsymbol{s}_b + (1-\gamma)\boldsymbol{s}_c, \quad (10)$$

where $\mathbf{W}$ are the learnable weight matrices, σ is the sigmoid activation function, and γ represents the gating probability. This adaptive mechanism controls the influence of each representation on the final user preference representation $\boldsymbol{s}_u$. The recommendation score for each item is computed using dot-product similarity as:

$$P_{rec}(i) = \text{softmax}(\boldsymbol{s}_u \cdot \boldsymbol{I}_i), \quad (11)$$

where $\boldsymbol{I}_i$ are the item representations[2]. Finally, we optimize the recommendation loss $\mathcal{L}_{\text{rec}}$ as follows:

$$\mathcal{L}_{\text{rec}} = -\sum_{j=1}^{N}\sum_{i=1}^{M} y_{ij} \log\left(\boldsymbol{P}_{\text{rec}}^{(j)}(i)\right), \quad (12)$$

where N is the number of conversations, M is the number of items, and y_{ij} is the ground truth label indicating whether item i is relevant to conversation j.

V. Experiments

A. Experimental Settings

1) Datasets: We conduct our experiments on two widely used English CRS datasets: ReDial [4] and INSPIRED [40] datasets. ReDial dataset, focused on movie recommendations, contains 11,348 conversations and is constructed through crowdsourcing on Amazon Mechanical Turk (AMT). The INSPIRED dataset contains 999 conversations, also about movie recommendations, but additionally provides recommendation strategies based on social science theories. All experiments are conducted using the official dataset splits. We constructed the KG by scraping data from IMDB[3], using movie titles and release years as key search terms.

2) Baselines: We consider a comprehensive range of baseline models, including traditional CRS **ReDial** [4], knowledge-graph based methods such as **KBRD** [7] and **KGSF** [5], language model-based methods like **BERT** [35], **GPT-2** [36], and **Llama3.1-8B** [11], metadata-augmented language model **PECRS** [37], and hybrid approaches that combine KGs with language models: **BARCOR** [24] fuses BART with a KG, **UniCRS** [8] employs KG prompt learning, **DCRS** [38] leverages retrieval-augmented prompt learning, and **MSCRS** [39] introduces multi-modal graph prompt learning. Since no existing baselines specifically generate user preference summaries, we use **Llama3.1-8B** and **GPT-4o** as baselines for comparison, both without KG access. For clarity, we denote **Llama3.1-8B** as **Llama-Summary** when used for preference summary generation.

3) Evaluation Metrics: Our evaluation framework assesses both the performance of the recommendations and the quality of generated user preference summaries. For recommendation tasks, we adopt widely used metrics from previous works [5], [23], including HR@K, NDCG@K, and MRR@K, with K set to 10 and 50. To evaluate user preference summaries, we employ three complementary metrics: (1) Lexical Similarity: ROUGE-1, ROUGE-2, ROUGE-L, and ROUGE-Sum; (2) Semantic Understanding: Following [41], we use GPT-4o-mini to assess Reasoning Proficiency (RP) and Factual Consistency (FC). RP assesses logical coherence, inference accuracy, and

[1]The preference representation $\boldsymbol{s}_b$ is specific to each CRS model as different models employ distinct methods for building user preferences.

[2]The item representations $\boldsymbol{I}_i$ are specific to each CRS model and may vary depending on the architecture employed by the base CRS.

[3]https://www.imdb.com/

TABLE I: Performance comparison on recommendation tasks. 'COM.' denotes models enhanced with the COMPASS approach and 'Improv.' indicates the relative improvement of COMPASS compared to the original base model. The best results are highlighted in bold (t-test with $p < 0.05$).

Types	Model	ReDial						INSPIRED					
		HR@10	HR@50	NDCG@10	NDCG@50	MRR@10	MRR@50	HR@10	HR@50	NDCG@10	NDCG@50	MRR@10	MRR@50
Base	ReDial [4]	0.140	0.320	0.061	0.065	0.035	0.045	0.117	0.285	0.035	0.072	0.022	0.048
	BERT [35]	0.143	0.319	0.073	0.108	0.052	0.059	0.179	0.328	0.095	0.125	0.072	0.079
	GPT-2 [36]	0.147	0.327	0.071	0.107	0.051	0.056	0.112	0.278	0.089	0.128	0.063	0.076
	Llama3.1-8B [11]	0.188	0.376	0.103	0.146	0.078	0.087	0.190	0.332	0.118	0.150	0.094	0.102
	KBRD [7]	0.151	0.336	0.099	0.136	0.071	0.079	0.172	0.265	0.106	0.127	0.086	0.091
	KGSF [5]	0.183	0.378	0.098	0.140	0.072	0.081	0.175	0.273	0.106	0.128	0.088	0.093
	PECRS [37]	0.205	0.399	0.112	0.154	0.083	0.093	0.179	0.337	0.106	0.142	0.084	0.092
	BARCOR [24]	0.169	0.374	0.088	0.133	0.063	0.073	0.185	0.339	0.104	0.137	0.080	0.087
	UniCRS [8]	0.216	0.416	0.118	0.163	0.087	0.095	0.250	0.408	0.182	0.218	0.148	0.160
	DCRS [38]	0.242	0.422	0.142	0.182	0.111	0.120	0.225	0.417	0.152	0.194	0.128	0.135
	MSCRS [39]	0.255	0.448	0.153	0.192	0.122	0.130	0.257	0.424	0.163	0.205	0.145	0.151
Enhanced	COM.+KBRD	0.199	0.412	0.103	0.150	0.075	0.085	0.249	0.392	0.152	0.183	0.123	0.129
	+Improv.	31.79%	22.61%	4.04%	10.29%	5.63%	7.59%	44.76%	47.92%	43.39%	44.09%	43.02%	38.70%
	COM.+KGSF	0.198	0.413	0.105	0.152	0.076	0.088	0.197	0.400	0.125	0.167	0.103	0.110
	+Improv.	8.20%	9.26%	7.14%	8.57%	5.5%	8.64%	12.57%	46.52%	17.92%	30.47%	17.05%	18.28%
	COM.+BERT	0.182	0.382	0.098	0.142	0.073	0.082	0.207	0.345	0.129	0.157	0.105	0.110
	+Improv.	27.27%	19.75%	34.25%	31.48%	40.38%	38.98%	15.64%	5.18%	35.79%	25.60%	45.83%	39.24%
	COM.+Llama3.1-8B	0.215	0.406	0.118	0.161	0.089	0.100	0.232	0.377	0.146	0.169	0.117	0.122
	+Improv.	13.76%	7.98%	14.45%	14.10%	12.82%	14.94%	22.11%	13.55%	23.73%	12.67%	24.47%	19.61%
	COM.+MSCRS	**0.264**	**0.451**	**0.160**	**0.200**	**0.129**	**0.137**	**0.277**	**0.427**	**0.183**	**0.214**	**0.154**	**0.160**
	+Improv.	3.53%	0.67%	4.58%	4.17%	5.74%	5.38%	7.78%	0.71%	12.27%	4.39%	6.21%	5.96%

recommendation relevance, while FC evaluates dialogue alignment, knowledge graph consistency, and statement accuracy. Both metrics are scored on a scale of 0 to 100. (3) User-Centric Evaluation: Inspired by [42], [43], we conduct LLM-based simulated user evaluations, which rank the system's output based on two aspects: Explainability, which measures how clearly the system justifies its recommendations, and User Preference Alignment, which measures how accurately the system understands and responds to user preferences expressed in the conversation.

4) Implementation Details: We implement COMPASS using the Llama3.1-8B model [11] as the base LLM, freeze all base LLM parameters and employ LoRA [44] for efficient fine-tuning. The graph encoder uses 1 layer with 768 hidden dimensions. For entity linking, we use provided entities for INSPIRED dataset and GPT-3.5 to extract entities from Redial dialogues, then apply text-embedding-ada-002 for semantic matching to KG entities. We use batch sizes of 256 for pre-training and 128 for fine-tuning on Inspired dataset. For ReDial dataset, we maintain a batch size of 128 for both pre-training and fine-tuning. All experiments run on Nvidia A100 GPUs. For ground truth preference summaries, we employ GPT-4 to generate summaries based on dialogue histories and associated item metadata from the KG using structured instruction templates[4].

B. Recommendation Evaluation

1) Improvement over Baseline Models: COMPASS is designed to be flexible, allowing integration with different CRS models. We evaluated its effectiveness across various models, with performance results shown in Table I. We have the following observations: (1) Baseline models incorporating external KGs, such as KBRD and KGSF, consistently outperform simpler language model-based approaches like BERT and GPT-2. This highlights the importance of structured knowledge in capturing user preferences and item relationships. settings. (2) Llama3.1-8B demonstrates strong performance, surpassing BARCOR and performing competitively with PECRS on ReDial dataset, indicating that the advanced language understanding and extensive world knowledge of LLMs can effectively capture user preferences, resulting in more accurate recommendations. (3) Advanced hybrid methods that integrate language models with KGs achieve the strongest baseline performance, with MSCRS achieving the best results across all metrics on both datasets.

Building upon these baselines, COMPASS consistently enhances every integrated model regardless of architecture or performance level. The framework delivers substantial gains on weaker models, achieving 44.76% HR@10 improvement with KBRD on INSPIRED dataset and 27.27% with BERT on ReDial dataset. When integrated with LLMs like Llama3.1-8B, COMPASS maintains solid enhancements of 13.76% in HR@10 on ReDial dataset. Most importantly, COMPASS enhances the strongest baseline MSCRS, achieving 7.78% HR@10 improvement and reaching the best overall performance across all metrics. These consistent improvements across different architectures validate that COMPASS's knowledge-enriched preference representations provide universal value. These representations effectively capture user intent while providing structured insights that enhance both recommendation performance and explainability.

2) Comparison of Enhancement Methods: To evaluate COMPASS's effectiveness, we compare it with Llama-Summary and GPT-4o as baseline enhancers, which generate user preference summaries without KG access. Figure 3 illustrates the performance of these methods across various CRSs on INSPIRED dataset. While all enhancement methods improve upon base models, COMPASS consistently

[4]Code, data, and instruction templates are available at: https://github.com/icedpanda/COMPASS-official

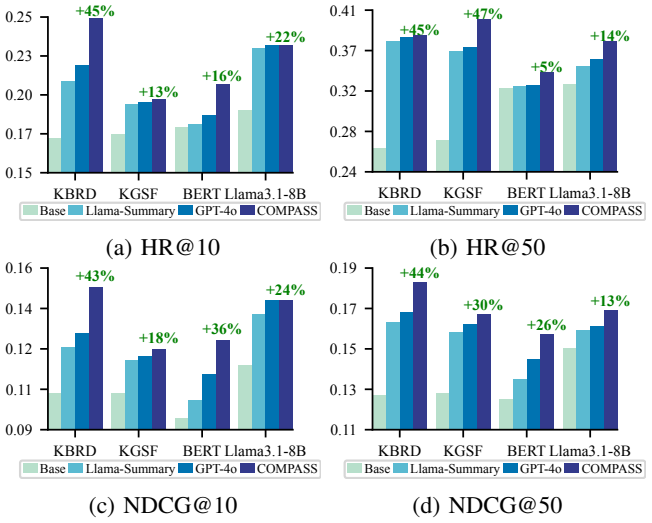


Fig. 3: Comparison of recommendation performance on the INSPIRED dataset with different enhancers. Green percentages show improvements over baselines.

outperforms both alternatives. Notably, COMPASS surpasses GPT-4o despite the latter being a much larger model, with improvements ranging from marginal gains on Llama3.1-8B to substantial improvements on KBRD and KGSF. This superior performance demonstrates the critical importance of integrating domain-specific knowledge into preference modeling. The substantial performance gap between COMPASS and Llama-Summary highlights the effectiveness of our framework in leveraging both structured knowledge and LLM capabilities. These results show COMPASS's ability to generate more accurate and contextually relevant user preference summaries, leading to improved recommendation performance across different CRS architectures.

3) Ablation study: To assess the contribution of different components in COMPASS, we conducted an ablation study by evaluating several model variants across three base models: KBRD, KGSF, and LLaMA3.1-8B. We compare the COMPASS model against three ablated versions: (1) **COM w/o REC**: The generated preference summary does not include recommended items; (2) **COM w/o GEP**: COMPASS without the graph entity captioning pre-training; (3) **COM w/o KG**: COMPASS without the graph encoder.

The results in Table II show that COMPASS generally achieves the best performance across most metrics, highlighting its superior capability. The COM w/o REC variant shows the largest performance drop, emphasizing the crucial role of including recommended items in preference summaries to guide downstream models. Both COM w/o KG and COM w/o GEP variants demonstrate comparable performance declines, highlighting the crucial roles of KG integration for understanding item relationships and GEP for comprehending KG-augmented context. These findings show the importance of each COMPASS component, with their synergy driving the superior performance of the full model.

TABLE II: Ablation study on recommendation task performance. The best-performing results are highlighted in bold.

Model	ReDial					
	HR@10	HR@50	NDCG@10	NDCG@50	MRR@10	MRR@50
KBRD	0.151	0.336	0.099	0.136	0.071	0.079
+COM w/o KG	0.195	0.405	0.102	0.147	0.074	0.084
+COM w/o REC	0.190	0.395	0.100	0.146	0.075	0.083
+COM w/o GEP	0.188	0.410	0.100	0.148	0.072	0.082
+COMPASS	**0.199**	**0.412**	**0.103**	**0.150**	**0.075**	**0.085**
KGSF	0.183	0.378	0.098	0.140	0.072	0.081
+COM w/o KG	0.196	0.410	0.102	0.150	0.074	0.085
+COM w/o REC	0.192	0.408	0.101	0.150	0.074	0.084
+COM w/o GEP	0.196	0.406	0.102	0.149	0.073	0.084
+COMPASS	**0.198**	**0.413**	**0.105**	**0.152**	**0.076**	**0.088**
Llama3.1-8B	0.188	0.376	0.103	0.146	0.078	0.087
+COM w/o KG	0.209	0.399	0.116	0.159	0.087	0.096
+COM w/o REC	0.202	0.395	0.109	0.151	0.081	0.090
+COM w/o GEP	0.212	0.405	0.117	**0.162**	0.088	0.099
+COMPASS	**0.215**	**0.406**	**0.118**	0.161	**0.089**	**0.100**

TABLE III: Evaluation of generated user preference summaries. RP is reasoning proficiency and FC is factual consistency. The best results are highlighted in bold.

Model	ReDial					
	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Sum	RP	FC
COM w/o KG	62.06	39.16	49.91	59.33	81.33	82.13
COM w/o NP	61.72	38.95	49.74	59.00	81.16	82.54
COMPASS	**62.71**	**40.61**	**51.14**	**60.03**	**82.20**	**84.21**
Model	INSPIRED					
	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Sum	RP	FC
COM w/o KG	55.85	32.94	43.36	53.12	82.59	84.33
COM w/o NP	58.20	34.57	44.73	55.08	82.36	84.29
COMPASS	**59.37**	**35.93**	**46.88**	**56.64**	**83.14**	**85.17**

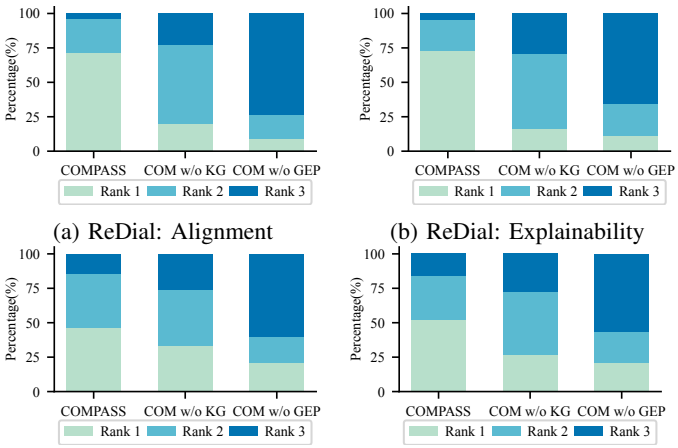
C. Preference Generation Evaluation

To evaluate COMPASS's ability to reason over user preferences and generate explainable summaries, we compare COMPASS with variants to assess key component effectiveness. Additionally, a case study with Llama-Summary and GPT-4o examples provides qualitative insights.

1) Automatic Evaluation: Table III presents a comparative analysis of COMPASS and its variants, COM w/o KG and COM w/o GEP, on ReDial and INSPIRED datasets. COMPASS consistently outperforms its variants across all metrics on both datasets. Lexical similarity metrics (ROUGE scores) demonstrate that COMPASS generates summaries that align closely with reference texts, while higher reasoning proficiency and factual consistency scores illustrate its superior reasoning quality and factual accuracy. The performance gap between COMPASS and COM w/o KG demonstrates the value of KG integration in enhancing reasoning over user preferences. Incorporating structured domain knowledge enables more accurate preference inference and explainable summary generation. Moreover, the comparative performance of COM w/o GEP and COMPASS indicates that, despite the presence of structured knowledge from the KG, there remains a modality gap between graph structures and natural language, posing challenges for LLMs to effectively reason with KG information. These results validate the COMPASS framework, demonstrating the synergistic effect of combining KGs with LLMs and confirming the critical role of our graph

TABLE IV: Case Study: Comparison of preference summaries generated by different models. Green indicates information supported by the KG, yellow indicates correct inferences not explicitly detailed in the KG but aligned with the content, and red indicates unsupported and unreasonable details.

Hist.	**System**: Hello! What kind of movies are you into? **User**: I loved *The Professional (1981)* and would love to see something similar.
User Preference Summary	**COMPASS:** *Reasoning:* The user mentioned 'The Professional (1981)' as a movie they loved and would like to see something similar. By analyzing the knowledge graph, we can identify that 'The Professional' falls under the genres of Action, Crime, and Thriller, and includes themes such as professional killers, assassins, and intense action sequences. The movie also features notable actors like Paul Belmondo and Michel Beaune, and is directed by Georges Lautner. These elements suggest the user enjoys high-stakes, action-packed narratives with complex characters and moral dilemmas. *Overall preferences*: Action, Crime, Thriller, professional killers, assassins, intense action sequences, complex characters, moral dilemmas, Jean-Paul Belmondo, Georges Lautner. *Current interests*: Action, Crime, Thriller, professional killers, assassins, intense action sequences.
	GPT-4o: *Reasoning:* The user mentioned a specific movie, 'The Professional (1981)', which they loved and expressed interest in finding similar movies. This indicates a preference for films with similar themes, styles, or elements. *Overall preferences*: crime, drama, thriller, intense character development, complex relationships, suspense. *Current interests*: The Professional (1981), crime drama, thriller, intense character dynamics.
	Llama-Summary: *Reasoning*: The user's conversation history indicates a preference for 1980s French cinema, particularly from the Nouvelle Vague movement, with an emphasis on intelligent, stylish, and critically-acclaimed films that often involve morally complex situations. *Overall preferences*: French New Wave, Intellectual, Stylish, Morally Complex. *Current interests*: Jean-Pierre Mousel, The Professional (1981), Claude Sautet, Claudie Claira.
	COM w/o GEP: *Reasoning*: The user mentioned 'The Professional (1981)' as a favorite movie and expressed interest in finding something similar. By analyzing the knowledge graph, we can identify that 'The Professional' falls under the genres of Action, Crime, and Thriller, and includes themes such as police, political corruption, and organized crime. The movie also features notable actors like Ryan O'Neal and Charles Grodin, and is directed by Donald Shebib. These elements suggest the user enjoys intense, plot-driven narratives with a focus on crime and political intrigue. *Overall preferences*: Action, Thriller, police, political corruption, organized crime, Ryan O'Neal, Charles Grodin, Donald Shebib. *Current interests*: Thriller, police, political corruption, organized crime.



(a) ReDial: Alignment (b) ReDial: Explainability

(c) INSPIRED: Alignment (d) INSPIRED: Explainability

Fig. 4: LLM-simulated user rankings for User Preference Alignment and Explainability across ReDial and INSPIRED datasets.

entity captioning pre-training in enabling effective cross-modal reasoning for explainable preference generation.

2) LLM-Simulated User Evaluation: To assess user-perceived quality, we employ LLM-simulated user evaluations on explainability and user preference alignment. Figure 4 presents the results across ReDial and INSPIRED datasets. COMPASS is consistently preferred over COM w/o KG and COM w/o GEP in both metrics. On ReDial dataset, COMPASS is most preferred in approximately 75% of the cases for both alignment and explainability. Although the preference gap is smaller on INSPIRED, COMPASS maintains a clear advantage. Notably, COM w/o NP is consistently the least preferred, showing the importance of pre-training in generating meaningful preference summaries. These results validate COMPASS's effectiveness in reasoning over user preferences and generating user-friendly, explainable summaries that enhance both fidelity and transparency in CRSs.

3) Case Study: We present a detailed case study in Table IV to demonstrate how different models reason over user preferences and generate explainable summaries from sample dialogue. COMPASS demonstrates compelling effectiveness in capturing and reasoning about user preferences. It accurately identifies the movie genres and lists key cast members and the director of *The Professional (1981)*. Moreover, COMPASS infers relevant themes that align with the movie's content, such as *professional killers*, and *intense action sequences*. In contrast, other models show varying levels of accuracy and reasoning depth. (1) GPT-4o, without KG augmentation, provides a broader interpretation but less precise reasoning. While it correctly identifies some genres, it includes unsupported elements, illustrating that general LLM knowledge alone is insufficient for accurate preference reasoning in domain-specific contexts. (2) Llama-Summary demonstrates significant inaccuracies, incorrectly attributing the film to the *French New Wave movement* and listing incorrect personnel. This highlights the limitations of smaller LLMs fin complex reasoning tasks without specialized knowledge. (3) COMP w/o GEP shows improved accuracy in genre reasoning but struggle with factual accuracy about cast and crew, making plausible but unsupported inferences. These comparisons emphasize the crucial roles that KG integration and proper pre-training play in enabling COMPASS to perform more accurate reasoning over user preferences, generating explainable summaries that enhance transparency and trust in CRSs.

VI. Conclusion

In this paper, we introduce COMPASS, a novel framework that augments LLMs with KGs to reason over user preferences in CRSs. To address the modality gap between structured

knowledge and natural language, we propose a graph entity captioning that transforms KG structures into LLM-compatible representations. Through knowledge-aware instruction tuning, COMPASS becomes proficient in performing cross-modal reasoning, generating interpretable user preference summaries. COMPASS has been extensively evaluated as a plug-and-play enhancement for various CRSs across benchmark datasets. The results demonstrate its effectiveness in significantly improving both recommendation performance and explainability of these base models. The adaptive integration mechanism of COMPASS allows for seamless enhancement of diverse CRS architectures without structural modifications, showcasing its versatility and potential for widespread adoption in the field of CRSs.

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